

ELEC 2FH4 - Midterm 2

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Question 1 (25 points)

If

$$\mathbf{D} = (2 \sin \theta \sin \phi) \mathbf{a}_r + (\cos \theta \sin \phi) \mathbf{a}_\theta + (2 \cos \phi) \mathbf{a}_\phi \quad \text{nC/m}^2$$

find:

(a) the charge density at the point

$$(3, 45^\circ, 30^\circ)$$

(b) the electric flux through the surface

$$r = 3, 0 < \theta < 45^\circ, 0 < \phi < 90^\circ$$

$$\frac{\text{rad}}{2\pi} = \frac{\text{deg}}{360}$$

$$\text{rad} = 2\pi \cdot \text{deg} / 360$$

$$\begin{aligned} \frac{d}{dx}(\sin x) &= \cos x \\ \frac{d}{dx}(\cos x) &= -\sin x \\ \frac{d}{dx}(x^n) &= nx^{n-1} \\ \frac{d}{dx}\left(\frac{1}{x}\right) &= -\frac{1}{x^2} \end{aligned}$$

$$\begin{aligned} \int \sin x \, dx &= -\cos x \\ \int \cos x \, dx &= \sin x \\ \int x^n \, dx &= \frac{x^{n+1}}{n+1} \\ \int \frac{1}{x} \, dx &= \ln x \end{aligned}$$

$$\vec{D} = (2 \sin \theta \sin \phi) \vec{a}_r + (\cos \theta \sin \phi) \vec{a}_\theta + (2 \cos \phi) \vec{a}_\phi \quad \text{nC/m}^2$$

a) charge density @ $(3, 45^\circ, 30^\circ) = \left(3, \frac{\pi}{4}, \frac{\pi}{6}\right) = ?$

$$\nabla \cdot \mathbf{D} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 D_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta D_\theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (D_\phi)$$

$$= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 2 \sin \theta \sin \phi) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \cdot \cos \theta \sin \phi) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (2 \cos \phi)$$

$$= \frac{1}{r^2} (4 r \sin \theta \sin \phi) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} \left(\frac{1}{2} \sin(2\theta) \sin \phi \right) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (2 \cos \phi)$$

$$= \frac{1}{r} 4 \sin \theta \sin \phi + \frac{1}{r \sin \theta} \left(\frac{1}{2} \cos(2\theta) \cdot 2 \cdot \sin \phi \right) + \frac{1}{r \sin \theta} (-2 \sin \phi)$$

$$= \frac{1}{r} 4 \sin \theta \sin \phi + \frac{1}{r \sin \theta} \cos(2\theta) \sin \phi + \frac{-1}{r \sin \theta} 2 \sin \phi$$

$$= \frac{1}{3} \cdot 4 \cdot \sin\left(\frac{\pi}{4}\right) \sin\left(\frac{\pi}{6}\right) + \frac{-1}{3 \sin\left(\frac{\pi}{4}\right)} \cos\left(2 \cdot \frac{\pi}{4}\right) \sin\left(\frac{\pi}{6}\right) + \frac{-1}{3 \sin\left(\frac{\pi}{4}\right)} 2 \sin\left(\frac{\pi}{6}\right)$$

$$= 0.471 + 0 - 0.471$$

$$= 0$$

$$\rho_v = 0$$

$$\sin \theta \cos \theta = \frac{1}{2} \sin(2\theta)$$

oops, i differentiated wrong...

retrying from this step below

$$= \frac{1}{r^2} (4 r \sin \theta \sin \phi) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} \left(\frac{1}{2} \sin(2\theta) \sin \phi \right) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (2 \cos \phi)$$

$$= \frac{1}{r} 4 \sin \theta \sin \phi + \frac{1}{r \sin \theta} \left(\frac{1}{2} \cos(2\theta) \cdot 2 \right) \sin \phi + \frac{-1}{r \sin \theta} 2 \sin \phi$$

$$= \frac{1}{r} 4 \sin \theta \sin \phi + \frac{1}{r \sin \theta} \cos(2\theta) \sin \phi + \frac{-1}{r \sin \theta} 2 \sin \phi$$

$$= \frac{1}{3} \cdot 4 \cdot \sin\left(\frac{\pi}{4}\right) \sin\left(\frac{\pi}{6}\right) + \frac{1}{3 \sin\left(\frac{\pi}{4}\right)} \cos\left(2 \cdot \frac{\pi}{4}\right) \sin\left(\frac{\pi}{6}\right) + \frac{-1}{3 \sin\left(\frac{\pi}{4}\right)} 2 \sin\left(\frac{\pi}{6}\right)$$

→ double checking if they actually cancel out

$$= 0.471 + 0 - 0.471$$

$$= 0$$

$$= \rho_v \longrightarrow \rho_v = 0$$

$$b) \quad r = 3, \quad 0 < \theta < 45^\circ, \quad 0 < \phi < 90^\circ \\
0 < \theta < \pi/4, \quad 0 < \phi < \pi/2$$

$$\Phi = \int_S \mathbf{D} \cdot d\mathbf{s}$$

$$= \iint \mathbf{D}_r \cdot r^2 \sin \theta \, d\theta \, d\phi \, \hat{\mathbf{a}}_r$$

$$= \iint 2 \sin \theta \sin \phi \, r^2 \sin \theta \, d\theta \, d\phi$$

$$= 2r^2 \cdot \int_0^{\pi/4} (\sin \theta)^2 \, d\theta \cdot \int_0^{\pi/2} \sin \phi \, d\phi$$

$$= 2r^2 \cdot \left[\frac{\theta}{2} - \frac{\sin(2\theta)}{4} \right] \Big|_0^{\pi/4} \cdot \left[-\cos \phi \right] \Big|_0^{\pi/2} \Big|_{r=3}$$

$$= 2(3)^2 \cdot 0.1427 \cdot 1$$

$$= 2.5686 \, \text{nC}$$

$$\Phi = 2.569 \, \text{nC}$$

Question 2 (25 points)

Find the energy stored in the spherical region

$$r \leq 3 \text{ m}, 0 < \theta < \frac{\pi}{2}, 0 < \phi < \pi$$

where, $\mathbf{E} = 3r \sin \theta \cos \phi \mathbf{a}_r + 2r \cos \theta \cos \phi \mathbf{a}_\theta - r \sin \phi \mathbf{a}_\phi \text{ V/m}$, exists.

- $\vec{E} = 3r \sin \theta \cos \phi \mathbf{a}_r + 2r \cos \theta \cos \phi \mathbf{a}_\theta - r \sin \phi \mathbf{a}_\phi \text{ V/m}$
- $0 \leq r \leq 3 \text{ m}$
- $0 < \theta < \pi/2$
- $0 < \phi < \pi$
- $\epsilon \text{ stored in region} = ?$

$$|\mathbf{E}|^2 = \left| 3r \sin \theta \cos \phi \mathbf{a}_r + 2r \cos \theta \cos \phi \mathbf{a}_\theta - r \sin \phi \mathbf{a}_\phi \right|^2$$

$$= 3^2 r^2 \sin^2 \theta \cos^2 \phi + 2^2 r^2 \cos^2 \theta \cos^2 \phi + r^2 \sin^2 \phi$$

$$= 9 r^2 \sin^2 \theta \cos^2 \phi + 4 r^2 \cos^2 \theta \cos^2 \phi + r^2 \sin^2 \phi$$

$$W = \frac{\epsilon_0}{2} \int_V |\mathbf{E}|^2 dV, \quad dV = r^2 \sin \theta dr d\theta d\phi$$

$$= \frac{\epsilon_0}{2} \iiint (9 r^2 \sin^2 \theta \cos^2 \phi + 4 r^2 \cos^2 \theta \cos^2 \phi + r^2 \sin^2 \phi) \cdot r^2 \sin \theta dr d\theta d\phi$$

$$= \frac{\epsilon_0}{2} \cdot \int_0^3 r^4 dr \cdot \int \int (9 \sin^3 \theta \cos^2 \phi + 4 \cos^2 \theta \sin \theta \cos^2 \phi + \sin^2 \phi \sin \theta) d\theta d\phi$$

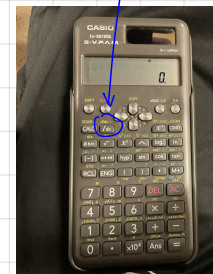
$$= \frac{\epsilon_0}{2} \cdot \int_0^3 r^4 dr \cdot \left[9 \int_0^{\pi/2} \sin^3 \theta d\theta \int_0^\pi \cos^2 \phi d\phi + 4 \int_0^{\pi/2} \cos^2 \theta \sin \theta d\theta \int_0^\pi \cos^2 \phi d\phi + \int_0^{\pi/2} \sin \theta d\theta \int_0^\pi \sin^2 \phi d\phi \right]$$

$$= \frac{\epsilon_0}{2} \cdot \frac{1}{5} r^5 \Big|_0^3 \cdot \left[9 \cdot 0.667 \cdot 1.571 + 4 \cdot 0.333 \cdot 1.571 + 1 \cdot 1.571 \right]$$

$$= \frac{\epsilon_0}{2} \cdot \frac{1}{5} (3)^5 \cdot (13.094)$$

$$W = 2.817 \times 10^{-9} \text{ J}$$

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Question 3 (50 points)

Consider the cylindrical surface

$$\rho = 3, 0 < \phi < \frac{\pi}{2}, 0 < z < 4 \text{ m.}$$

(a) The current density in the region is given by

$$\mathbf{J} = \frac{12}{\rho} \cos \phi \mathbf{a}_\rho \text{ A/m}^2.$$

Determine the total current flowing outward through the cylindrical surface.

(b) In a dielectric material with relative permittivity

$$\epsilon_r = 4,$$

the polarization vector is $\mathbf{P} = \frac{150}{\rho} \mathbf{a}_\rho \text{ nC/m}^2$. Find the electric field $\mathbf{E}$ and the electric flux density $\mathbf{D}$ at $\rho = 3 \text{ m}$.

- $\rho = 3$
- $0 < \phi < \pi/2$
- $0 < z < 4 \text{ m}$

a) $\mathbf{J} = 12 \frac{1}{\rho} \cos \phi \mathbf{a}_\rho \text{ A/m}^2$
 $I = ?$

$$I = \int_S \mathbf{J} \cdot d\mathbf{S}, \quad d\mathbf{S} = \rho d\rho d\phi dz$$

$$= \iiint 12 \frac{1}{\rho} \cos \phi \cdot \rho d\rho d\phi dz \quad | \rho = 3$$

$$= 12 \cdot \int_0^{\pi/2} \cos \phi d\phi \cdot \int_0^4 dz$$

$$= 12 \cdot \sin \phi \Big|_0^{\pi/2} \cdot z \Big|_0^4$$

$$= 12 \cdot 1 \cdot 4$$

$$I = 48 \text{ A}$$

b) $\epsilon_r = 4$
 $\mathbf{P} = 150 \text{ n} \cdot \frac{1}{\rho} \mathbf{a}_\rho \text{ C/m}^2$
 $\vec{E} = ?$
 $\vec{D} = ?$
 @ $\rho = 3$

$$P = \chi_e \epsilon_0 E$$

$$E = \frac{P}{\chi_e \epsilon_0}$$

$$= \frac{150 \text{ n} \cdot \frac{1}{\rho}}{2.5 \epsilon_0} \mathbf{a}_\rho$$

$$= \frac{150 \text{ n}}{2.5 \epsilon_0 (3)} \mathbf{a}_\rho$$

$$= 2'258.818 \mathbf{a}_\rho$$

$$\vec{E} = 2.259 \text{ k} \mathbf{a}_\rho$$

$$D = \epsilon_0 \epsilon_r E$$

$$= \epsilon_0 (4) (2.259 \text{ k} \mathbf{a}_\rho)$$

$$= 8.000644 \times 10^{-8} \mathbf{a}_\rho$$

$$\vec{D} = 8.001 \times 10^{-8} \mathbf{a}_\rho$$